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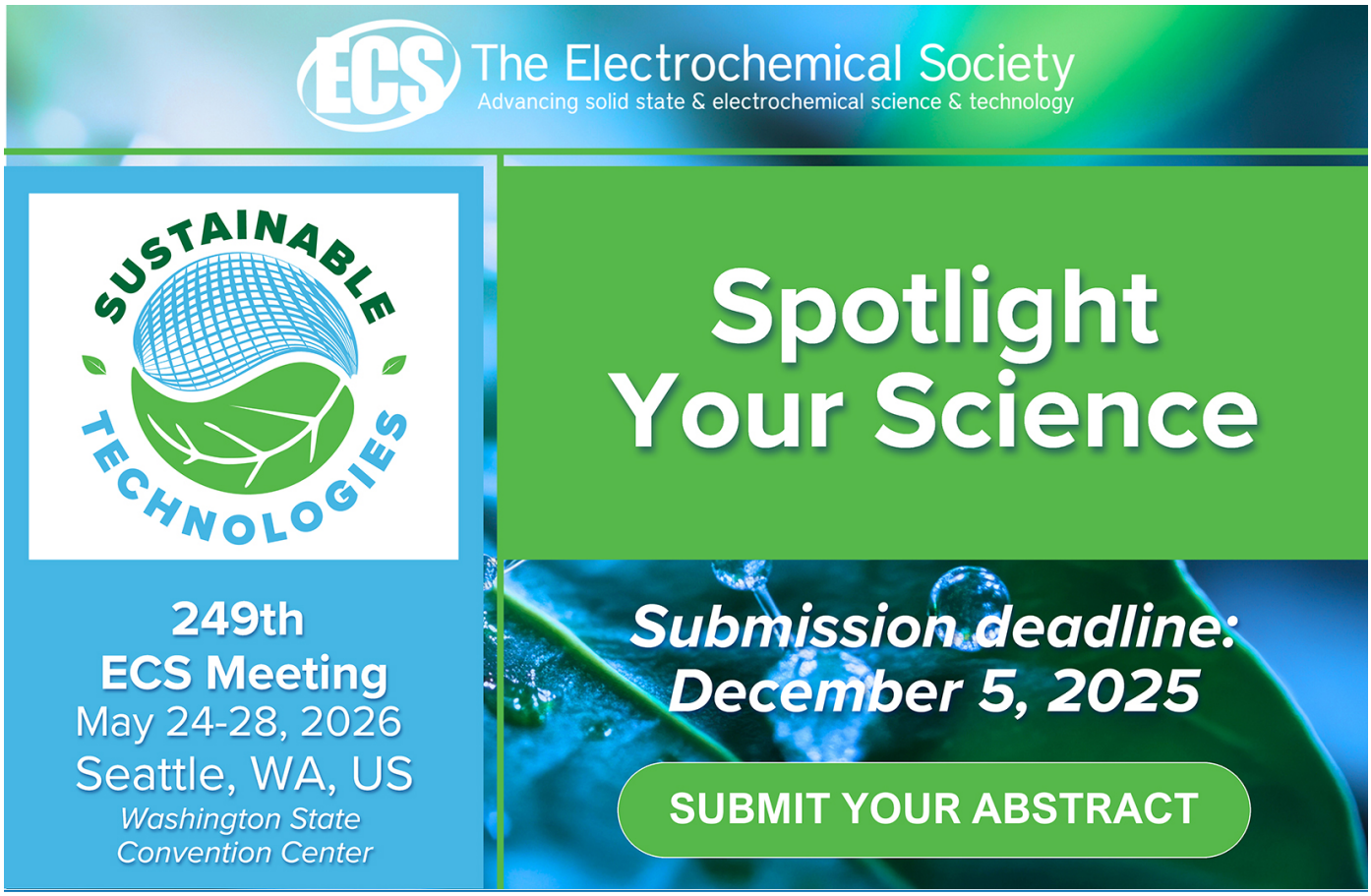

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Research on Obstacle Avoidance Path Planning for Robotic Arms Using Improved RRT Algorithm

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Abstract. In robot path planning, the Rapidly-exploring Random Tree (RRT) algorithm is extensively employed for its effective search performance. However, the conventional RRT algorithm exhibits several limitations, including non-smooth paths, low search efficiency, and insufficient adaptability to dynamic environments. To tackle these issues, this paper introduces an improved RRT algorithm that integrates a target-biased sampling strategy, bidirectional expansion, adaptive step size adjustment, multi-objective optimization, and path smoothing. The target-biased sampling strategy dynamically adjusts the sampling probability, enhancing global exploration capability and accelerating convergence speed. The bidirectional expansion mechanism enables the search tree to expand concurrently from both the start and goal points, thereby substantially decreasing the search time. Adaptive step size adjustment modifies the step size based on the distance to the goal, improving the accuracy of path planning. Additionally, multi-objective optimization, combined with genetic algorithms, optimizes path length, smoothness, and motion time. Path smoothing is achieved through the application of cubic B-splines, which optimize path continuity and minimize unnecessary sharp turns. The experimental outcomes indicate that the enhanced algorithm surpasses conventional RRT and RRT* algorithms in search efficiency, path quality, and computational performance. This makes it a more effective approach for robot path planning in complex and dynamic environments.

1. Introduction

Path planning is a crucial issue in robotics and autonomous systems, particularly in complex environments. The Rapidly-exploring Random Tree (RRT) algorithm is widely recognized for its efficiency[1], but it suffers from issues like high randomness, non-smooth paths, and slow convergence, which hinder its performance, particularly in cases where both efficiency and precision are required. While the RRT* algorithm guarantees path quality through asymptotic optimality[2], it still faces challenges in real-time applications, especially in dynamic environments, due to high computational complexity and slow response to dynamic obstacles[3].

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To overcome these limitations, this paper introduces an enhanced RRT algorithm that features several important innovations. The proposed approach integrates a target-biased sampling strategy, a bidirectional expansion mechanism, adaptive step size adjustment, multi-objective optimization, and path smoothing. The target-biased sampling strategy enhances exploration efficiency in the target region, while the bidirectional expansion mechanism speeds up the search process by growing trees from both the start and goal points simultaneously, adaptive step size adjustment optimizes exploration by dynamically modifying the step size according to the distance to the goal. Multi-objective optimization, implemented using genetic algorithms, balances path length, smoothness, and motion time, resulting in more practical and efficient paths. Additionally, path smoothing, achieved through cubic B-splines, eliminates redundant nodes and improves path continuity.

2. Related Works

Since its inception, the Rapidly-exploring Random Tree (RRT) algorithm and its variants have found extensive application in robot path planning, especially for autonomous obstacle avoidance in both static and dynamic environments. In their pioneering 2000 paper, S. M. LaValle and J. J. Kuffner laid the theoretical groundwork for the RRT algorithm.[1]. Subsequently, S. Karaman and E. Frazzoli proposed the RRT* algorithm, which demonstrates exceptional performance in high-dimensional spaces[2]. J. D. Gammell and colleagues further enhanced the field by presenting the Informed RRT* algorithm, enhancing search efficiency through the use of elliptical heuristic functions[4].

Yin et al[5]. proposed an enhanced Informed-RRT* method for robotic arm grasping planning, while Li et al.[6] enhanced the RRT algorithm to boost the obstacle avoidance capabilities of robotic arms, and Wan et al.[7] further optimized this process, achieving higher efficiency and reliability. Wu et al.[8] explored efficient path planning methods to increase the operational efficiency of robotic arms. Salzman O. et al.[9] presented the Asymptotically Optimal RRT algorithm, which notably enhanced both the speed and quality of motion planning. Additionally, M. Otte et al. proposed the RRTX algorithm, which supports rapid replanning in dynamic environments[10]. The Bi-AM-RRTStar algorithm, combining bidirectional search with attention mechanisms, demonstrated superior efficiency in dynamic environments[11]. Furthermore, Chen et al.[12] They introduced an enhanced path planning approach based on RRT-Connect, which overcomes the limitations of the traditional RRT-Connect algorithm and improves the robots' path planning abilities in dynamic environments.

Despite significant progress, most existing methods primarily focus on single-objective optimization, often neglecting the comprehensive optimization of path smoothness, motion time, and real-time replanning capabilities. The enhanced RRT algorithm introduced in this paper addresses these limitations through adaptive node selection, dynamic obstacle avoidance, multi-objective optimization, and path smoothing, thereby improving the efficiency and quality of path planning.

3. The Proposed Method

The overall procedure of the enhanced RRT algorithm is as follows: First, sampling points are generated using a goal-biased sampling strategy. Subsequently, a bidirectional expansion mechanism is employed, growing the path tree simultaneously from both the start and goal points. Next, an adaptive step-size adjustment strategy dynamically alters the step size. If no collision is detected, the step size is further adjusted to facilitate continuous growth; otherwise, the initial step size is retained for expansion. Once the two paths are connected to form an initial path, a multi-objective optimization strategy is applied for further refinement. Finally, redundant nodes are removed, and the path is smoothed using B-spline curves to obtain the final optimized path. The overall algorithmic flow is illustrated in Figure 1.

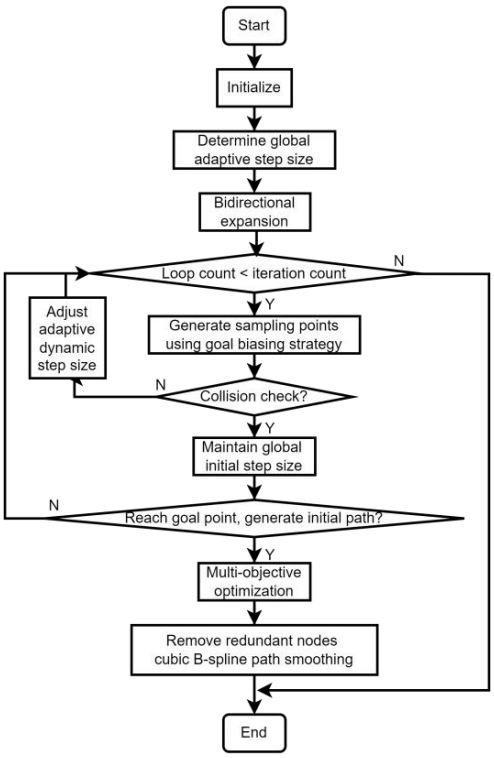


Figure 1. Improved RRT algorithm flowchart

3.1. Target Sampling Strategy Optimization

In the conventional RRT algorithm, sampling points are produced randomly, often leading to low search efficiency, especially in complex environments. The generated paths typically contain numerous redundant nodes, increasing path length and reducing smoothness. To boost the algorithm's performance for path planning in complex environments, this paper proposes an improved target-biased sampling strategy.

The core of this strategy is to increase the tendency toward the goal direction by introducing a target-biased probability parameter P_{bias} to control the selection of sampling points. Specifically, in each sampling iteration, the system produces a random number. If this number is less than or equal to the predefined P_{bias}, the goal point is chosen as the current sampling point, and the step size is enlarged. Otherwise, a point is randomly selected from the free space, and its validity is checked. If the point is valid, the predefined step size is used.

Furthermore, to further optimize this strategy, the value of P_{bias} can be dynamically modified according to the sampling success rate. The definition of P_{bias} is as follows:

$$P_{bias} = P_{min} + (P_{max} - P_{min}) \cdot exp\left(-\frac{k}{(T+1)^2}\right) \tag{1}$$

Here, P_{min} and P_{max} represent the minimum and maximum sampling probabilities, respectively, k is an environment complexity factor that controls the rate of probability adjustment, and T denotes the current sampling iteration. When the sampling success rate is high, P_{bias} is reduced to strengthen global exploration. Conversely, when the success rate is low, P_{bias} is increased to accelerate convergence toward the goal.

By incorporating the target-biased sampling strategy and the adaptive probability adjustment method, the algorithm's global exploration capability is significantly enhanced, improving search efficiency in the target region. Particularly in complex environments, this approach reduces the number of invalid samples, increases convergence speed, and improves the smoothness of the generated paths.

3.2. Bidirectional Expansion Mechanism

The traditional RRT algorithm expands unidirectionally from the start point toward the goal, which often results in low search efficiency, particularly in complex environments where the distance between the two points is large. To address this limitation, this paper introduces a bidirectional expansion mechanism, where the path tree expands concurrently from both the start and goal points. This method greatly decreases the number of iterations needed to find a feasible path, thereby enhancing overall performance.

In this method, two trees expand in parallel following standard RRT rules. During each iteration, new nodes are checked for collisions. If the two trees are sufficiently close and no obstacles are present between them, a feasible path is formed. This simultaneous exploration increases the likelihood of quickly finding a valid path while balancing global exploration with local refinement. By minimizing unnecessary expansions, the bidirectional mechanism improves path planning efficiency and generates smoother, more practical paths, especially in complex environments.

3.3. Adaptive Step Size Adjustment

In traditional RRT-based path planning, a fixed step size is typically used, which significantly impacts performance. A large step size may cause the expansion to overshoot the target region or collide with obstacles, while a small step size reduces search efficiency and increases computation time. To tackle this problem, this paper introduces an adaptive step size adjustment strategy that dynamically alters the step size according to the distance between the current node and the goal, thereby improving both search efficiency and accuracy.

The fundamental concept of this method is to modify the step size based on the distance to the goal: a larger step size is used when the distance is large to rapidly approach the goal, while a smaller step size is employed when the distance is small to ensure precision and safety. The step size adjustment strategy is defined as follows:

$$s = \begin{cases} s_{min}, & d < d_{min} \\ s_{min} + \dfrac{d - d_{min}}{d_{max} - d_{min}} \cdot \left(s_{max} - s_{min}\right), & d_{min} \le d \le d_{max} \\ s_{max}, & d > d_{max} \end{cases} \tag{2}$$

Here, s represents the dynamically adjusted expansion step size; s_{min} and s_{max} denote the minimum and maximum step sizes, respectively; d represents the distance between the current node and the goal point, while d_{min} and d_{max} are the minimum and maximum distance thresholds. When the distance d between the node and the goal point is less than d_{min}, the step size is set to the minimum value s_{min} to ensure expansion accuracy. When d is larger than d_{max}, the step size is set to the maximum value s_{max} to accelerate the search. For intermediate distances, the step size is dynamically adjusted using linear interpolation.

The adaptive step size adjustment strategy enhances the flexibility of path planning by dynamically updating the expansion step size. When far from the target region, a larger step size accelerates the search and reduces the time spent on invalid expansions. As the algorithm approaches the target, the step size decreases to improve accuracy and prevent overshooting or collisions. Additionally, the strategy modifies the step size according to the environment of different nodes, further improving its adaptability to complex scenarios. This approach improves global efficiency while ensuring local precision, overcoming the limitations of fixed step sizes, and enhancing the robustness of the algorithm in various applications.

3.4. Multi-Objective Optimization

In path planning problems, optimizing only the path length often fails to meet practical application requirements, particularly in complex environments where path smoothness, obstacle avoidance, and kinematic characteristics are equally critical. To address this, this paper proposes a multi-objective optimization strategy that transforms the path planning problem into an optimization problem

considering multiple objectives, thereby generating smoother, shorter paths that meet practical application needs.

The multi-objective path optimization model incorporates multiple objective functions to comprehensively evaluate path quality. First, to ensure the path is as short as possible, thereby reducing motion time and energy consumption, the objective function for path length minimization is defined as:

$$f_1 = \sum_{i=1}^{N-1} \left\| P_i - P_{i+1} \right\| \tag{3}$$

Here, P_i and P_{i+1} represent adjacent nodes on the path, and $\|P_i - P_{i+1}\|$ denotes the Euclidean distance between them. Second, to reduce oscillations and acceleration changes in the robot's motion, the smoothness of the path is assessed using the second-order difference. The objective function for smoothness is formulated as:

$$f_2 = \sum_{i=2}^{N-1} \left\| P_i - \frac{P_{i-1} + P_{i+1}}{2} \right\| \tag{4}$$

Here, $\left\| P_i - \frac{P_{i-1} + P_{i+1}}{2} \right\|$ denotes the Euclidean distance between node P_i and the midpoint of its neighboring nodes P_{i-1} and P_{i+1}. Additionally, to ensure the path maintains a safe distance from obstacles, an obstacle distance function is introduced. The objective is to maximize the distance between each node in the path and the nearest obstacle. The objective function is defined as:

$$f_3 = -\min_{i=1,\ldots,N} d\left(P_i, O_j\right) \tag{5}$$

Here, $d(P_i, O_j)$ denotes the shortest distance between path node P_i and obstacle O_j, and the negative sign indicates the objective is to maximize this distance, thereby ensuring obstacle avoidance.

To achieve the aforementioned multi-objective optimization, this paper employs a Multi-Objective Genetic Algorithm (MO-GA) for global search and optimization. The algorithm first generates an initial path using the RRT algorithm, which is then trimmed and sampled to form the initial population. Next, for each path in the population, the values of the three objective functions—path length, smoothness, and obstacle avoidance—are calculated. Based on the Pareto optimal principle, the algorithm performs non-dominated sorting of the paths, retaining Pareto front solutions to ensure a balance among multiple objectives. Subsequently, genetic operations, including crossover and random perturbations (mutation) of path nodes, are applied to explore better solutions. The process iterates until the Pareto front of the objective functions stabilizes.

This multi-objective path optimization method not only achieves a balance between path length, smoothness, and obstacle avoidance but also significantly enhances the applicability and robustness of path planning.

3.5. Path Smoothing

The initial path generated by path planning often contains unnecessary sharp turns and redundant nodes due to the randomness of sampling and expansion, which negatively impacts both path length and smoothness. To address this issue, this paper employs a cubic B-spline-based smoothing method. First, redundant nodes are removed by traversing the path and eliminating nodes that do not contribute to the path's structure. Next, cubic B-splines are applied to smooth the path.

Cubic B-splines are effective interpolation methods that generate smooth, continuous curves while maintaining local control. Extra control points are inserted at the beginning and end of the path to ensure smooth transitions. The path is processed in segments, with each group consisting of four adjacent nodes generating a cubic B-spline curve. The mathematical expression of the cubic B-spline curve is as follows:

$$C(t) = \sum_{i=0}^{3} F_i(t) \cdot P_i \tag{6}$$

Here, P_i represents the control points of the path, and $F_i(t)$ denotes the cubic B-spline basis functions, given as:

$$\begin{cases} F_0(t) = \frac{1}{6}(1-t)^3, \\ F_1(t) = \frac{1}{6}(3t^3 - 6t^2 + 4), \\ F_2(t) = \frac{1}{6}(-3t^3 + 3t^2 + 3t + 1), \\ F_3(t) = \frac{1}{6}t^3 \end{cases} \tag{7}$$

Here, $t \in [0,1]$ represents the curve parameter. For each segment of the path, the corresponding cubic polynomial expression is calculated, and the path is smoothed by interpolating the path points. For example, for the *x, y, z* coordinates of the path, the fitting formulas are:

$$\begin{cases} x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3, \\ y(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3, \\ z(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3 \end{cases} \tag{8}$$

Here, the coefficients a_i, b_i, c_i are calculated based on the coordinates of the control points. Finally, all cubic B-spline segments are continuously concatenated to ensure the overall continuity and smoothness of the entire path.

By utilizing this cubic B-spline-based path smoothing method, the original path is converted from a series of discrete key points into a continuous and smooth curve. This method not only enhances the geometric continuity and smoothness of the path, eliminating sharp turns, but also effectively reduces the total path length, making it more compact. Additionally, the smoothed path significantly reduces oscillations and errors in the robot's motion, greatly enhancing kinematic performance and better meeting the requirements of practical motion planning.

4. Experiment and Results Analysis

4.1. Experimental Setup and Environment

To verify the performance and effectiveness of the enhanced RRT algorithm, simulation experiments were carried out using MATLAB R2019b. The hardware setup included a laptop running a 64-bit Windows 10 operating system, equipped with an Intel® Core™ i7-11700F processor and 16GB of RAM.

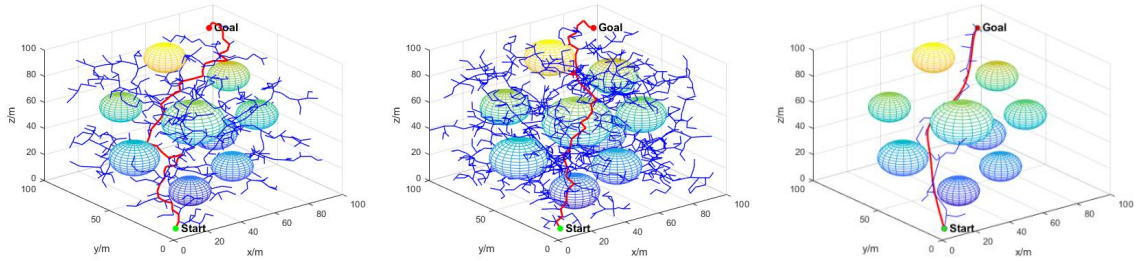
The experimental environment was set in a three-dimensional space with dimensions of 100×100×100. The starting point was set at (5, 5, 5), and the goal point was set at (90, 90, 90). Multiple spherical obstacles were placed in the environment to simulate a complex path planning scenario. The algorithm parameters were set as follows: the fixed iteration step size was 5, the target bias threshold ranged from 0.1 to 1, the connection threshold was 0.8, and the maximum number of iterations was 5,000. To ensure the reliability of the results, each algorithm was run independently 100 times, and the average values were calculated.

The comparison algorithms included the traditional RRT, RRT*, and the improved RRT. The traditional RRT served as the baseline, RRT* enhanced path quality through asymptotic optimality, and the improved RRT integrated a target-biased sampling strategy, bidirectional expansion, adaptive step size adjustment, path smoothing, and multi-objective optimization.

4.2. Experimental Results

Figure 2 illustrates the path planning results of the traditional RRT, RRT*, and improved RRT algorithms. In the figure, the circles represent obstacles, the blue lines depict the random expansion

process during algorithm execution, the green circle marks the starting point, the red circle marks the goal point, and the red solid line shows the final path generated by each algorithm. The path from the traditional RRT has many extra nodes, making it longer and less smooth. The RRT* algorithm makes the path better but still has some extra nodes and isn't very smooth. In contrast, the path from the improved RRT is much smoother, with a notable reduction in redundant nodes, resulting in a shorter and more efficient path.



a. Traditional RRT b. RRT* c. Improved RRT

Figure 2. The path search process diagram of the three algorithms

Table 1 provides a comparison of the traditional RRT, RRT*, and enhanced RRT algorithms in terms of search time, path length, number of path nodes, and number of sampled nodes. All metrics are averaged over 100 independent runs.

Table 1. The search results of the three algorithms

Methon	Search Time (s)	Average Length (m)	Average Path Nodes	Average Sampled Nodes
Traditional RRT	2.973	223.905	45.02	1311.957
RRT*	16.550	183.540	19.95	3118.060
Improved RRT	0.187	156.735	7.12	63.520

4.3. Result Analysis

From the data in Table 1, it is clear that the enhanced RRT algorithm substantially decreases the number of sampling points in 3D space compared to the traditional RRT and RRT* algorithms, while generating shorter, smoother paths with fewer bends. This decrease in sampling points is mainly due to the target-biased sampling strategy and adaptive probability adjustment, while the shorter, smoother paths result from the redundant node removal strategy and cubic B-spline-based path smoothing method.

The enhanced RRT algorithm shortens the path length by 30.0% compared to RRT and by 14.6% compared to RRT*. with redundant node elimination and path smoothing being the key contributors. Moreover, the search time is decreased by 93.7% compared to RRT and 98.9% compared to RRT*, owing to the target-biased sampling strategy and adaptive step size adjustment, which narrow the search space and improve efficiency. The average number of path nodes is cut by 84.2% compared to RRT and by 64.3% compared to RRT*, while the average number of sampled nodes is reduced by 95.2% compared to RRT and 98.0% compared to RRT*, demonstrating the efficiency improvements achieved through the target-biased sampling and adaptive step size approaches.

5. Conclusion

This paper introduces an enhanced RRT algorithm to overcome the limitations of traditional RRT algorithms, such as poor guidance, slow convergence, low search efficiency, and non-smooth paths. Key innovations, including target-biased sampling strategy optimization, bidirectional expansion, adaptive step size adjustment, path smoothing, and multi-objective optimization, significantly enhance path planning performance. The improved algorithm increases search efficiency by reducing

ineffective sampling and expansion time, balances global exploration and local optimization through adaptive step size adjustment, generates smoother paths using cubic B-splines, and produces more practical paths through multi-objective optimization. The experimental results show that the enhanced RRT algorithm surpasses the traditional RRT and RRT* algorithms in path length, search time, number of path nodes, and number of sampled nodes, effectively tackling path planning challenges in complex environments. This research provides a novel solution for efficient path planning with broad application prospects. Future work will explore its potential in robotic arm path planning and UAV navigation to further enhance its practicality and adaptability.

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